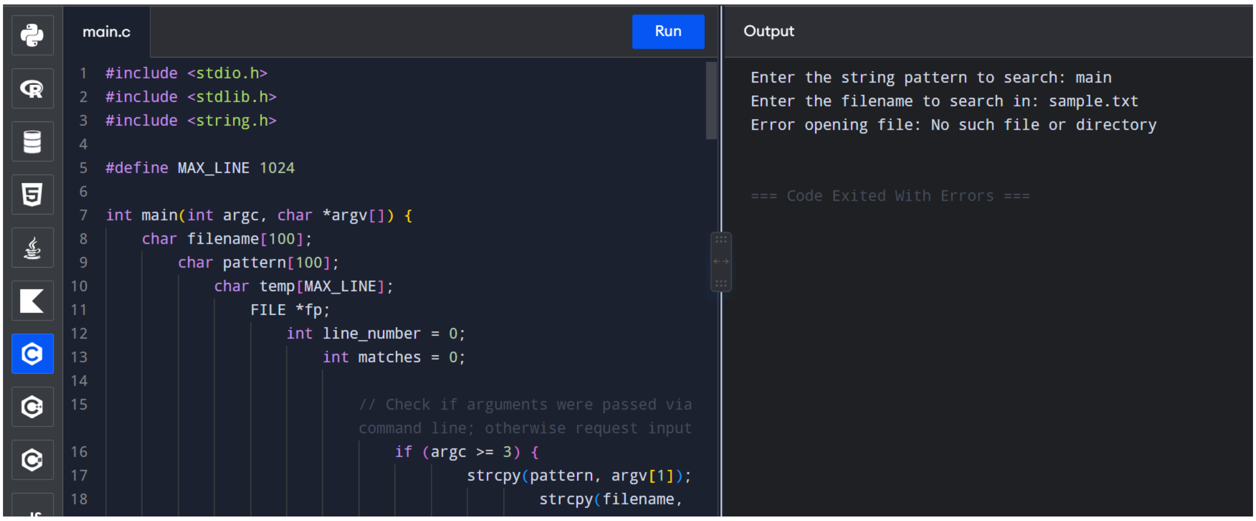
EXPERIMENT 38:

38. Write a C program for simulation of GREP UNIX command.



PSEUDO CODE :

- Get the target search string (pattern) and file path (filename) via command-line arguments or user prompt.
- Open the file using fopen() in read mode ("r"). If opening fails, print an error and terminate.
- Read the file line-by-line using fgets() inside a loop while maintaining a line_number counter.
- Use strstr() to check if pattern is present within the current line buffer.
- If strstr() returns a non-null pointer, print the line number and the matching line content.
- Close the file pointer using fclose().

CODE:

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>
```

```
#define MAX_LINE
1024


int main(int argc, char
*argv[]) {
    char filename[100];
    char pattern[100];
    char
temp[MAX_LINE];
    FILE *fp;
    int line_number = 0;
    int matches = 0;


    // Check if arguments
were passed via
command line;
otherwise request input
    if (argc >= 3) {
        strcpy(pattern,
```

```
argv[1]);

        strcpy(filename,
argv[2]);

    } else {

        printf("Enter the
string pattern to search:
");

        scanf("%s",
pattern);

        printf("Enter the
filename to search in: ");

        scanf("%s",
filename);

    }



    // Open target file

    fp = fopen(filename,
"r");

    if (fp == NULL) {
```

```
        perror("Error
opening file");
        return 1;
    }

    printf("\n--- Search
Results for '%s' in '%s' -
--\n", pattern, filename);

    // Read line by line
and search pattern
    while (fgets(temp,
MAX_LINE, fp) !=
NULL) {
        line_number++;
        if (strstr(temp,
pattern) != NULL) {
            printf("Line %d:
%s", line_number,
temp);
```

```
            matches++;
        }
    }

    if (matches == 0) {
        printf("No
matching occurrences
found.\n");
    } else {
        printf("\nTotal
matching lines found:
%d\n", matches);
    }

    fclose(fp);
    return 0;
}
```

INPUT :

```
#include <stdio.h>

int main() {

    printf("Inside main
function\n");

    return 0;

}
```

OUTPUT:

```
--- Search Results for
'main' in 'sample.txt' ---

Line 2: int main() {

Line 3:     printf("Inside
main function\n");



Total matching lines
found: 2
```